



天津师范大学

Tianjin Normal University

行列式的计算方法总结

张彪

天津师范大学

行列式的 计算方法



- ① 从定义出发
- ② 利用行列式性质
- ③ 按行或列展开计算
- ④ 箭头形行列式

- ⑤ 逐行逐列相加减
 - ⑥ 加边法
 - ⑦ 利用数学归纳法证明
 - ⑧ 利用递推公式
 - ⑨ 拆项法
- 

1.利用定义计算

$$\begin{vmatrix} 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & \cdots & 2 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & n-1 & \cdots & 0 & 0 \\ n & 0 & \cdots & 0 & 0 \end{vmatrix}$$

$$\begin{vmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 2 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & n-1 \\ n & 0 & 0 & \cdots & 0 \end{vmatrix}$$

2. 利用行列式性质把行列式化为上、下三角形行列式.

$$\begin{vmatrix} 3 & 1 & 1 & 1 \\ 1 & 3 & 1 & 1 \\ 1 & 1 & 3 & 1 \\ 1 & 1 & 1 & 3 \end{vmatrix} \quad \begin{vmatrix} 1+x & 1 & 1 & 1 \\ 1 & 1-x & 1 & 1 \\ 1 & 1 & 1+y & 1 \\ 1 & 1 & 1 & 1-y \end{vmatrix}$$

有时也可化成斜上、斜下三角形行列式

$$\begin{vmatrix} b & \cdots & b & b & a \\ b & \cdots & b & a & b \\ b & \cdots & a & b & b \\ \vdots & & \vdots & \vdots & \vdots \\ a & \cdots & b & b & b \end{vmatrix} = \begin{vmatrix} b & \cdots & b & b & a + (n-1)b \\ b & \cdots & b & a & a + (n-1)b \\ b & \cdots & a & b & a + (n-1)b \\ \vdots & & \vdots & \vdots & \vdots \\ a & \cdots & b & b & a + (n-1)b \end{vmatrix}$$

$$= (a + (n-1)b) \begin{vmatrix} b & \cdots & b & b & 1 \\ b & \cdots & b & a & 1 \\ b & \cdots & a & b & 1 \\ \vdots & & \vdots & \vdots & \vdots \\ a & \cdots & b & b & 1 \end{vmatrix} = (a + (n-1)b) \begin{vmatrix} b & \cdots & b & b & 1 \\ 0 & \cdots & 0 & a-b & 0 \\ 0 & \cdots & a-b & 0 & 0 \\ \vdots & & \vdots & \vdots & \vdots \\ a-b & \cdots & 0 & 0 & 0 \end{vmatrix}$$

$$= (-1)^{\frac{n(n-1)}{2}} (a + (n-1)b)(a-b)^{n-1}$$

3. 行列式按一行(一列)展开,或按多行(多列)展开(Laplace 定理)

例

设 $A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 3 & 4 \\ 1 & 1 & 2 & 3 \\ 2 & 3 & 4 & 4 \end{pmatrix}$, 计算 $A_{41} + A_{42} + A_{43} + A_{44}$ 和 $A_{14} + 3A_{24} + 2A_{34} + 4A_{44}$.

3. 行列式按一行(一列)展开,或按多行(多列)展开(Laplace 定理)

例

设 $A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 3 & 4 \\ 1 & 1 & 2 & 3 \\ 2 & 3 & 4 & 4 \end{pmatrix}$, 计算 $A_{41} + A_{42} + A_{43} + A_{44}$ 和 $A_{14} + 3A_{24} + 2A_{34} + 4A_{44}$.

解

$$A_{41} + A_{42} + A_{43} + A_{44} = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 3 & 4 \\ 1 & 1 & 2 & 3 \\ 1 & 1 & 1 & 1 \end{vmatrix} = 0, \quad A_{14} + 3A_{24} + 2A_{34} + 4A_{44} = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 3 & 3 \\ 1 & 1 & 2 & 2 \\ 2 & 3 & 4 & 4 \end{vmatrix} = 0.$$



$$D_{2n} = \begin{vmatrix} a_n & & & & b_n \\ & \ddots & & & \\ & & a_1 & b_1 & \\ & & c_1 & d_1 & \\ & \ddots & & & \\ c_n & & & & d_n \end{vmatrix}$$



$$D_{2n} = \begin{vmatrix} a_n & & & & & b_n \\ & \ddots & & & & \\ & & a_1 & b_1 & & \\ & & c_1 & d_1 & & \\ & & & & \ddots & \\ & & & & & d_n \\ c_n & & & & & \end{vmatrix} = \prod_{i=1}^n (a_i d_i - b_i c_i).$$

4. 箭头形行列式或者可以化为箭头形的行列式

设 $a_1, \dots, a_n \neq 0$.

$$\begin{vmatrix} a_0 & 1 & 1 & \cdots & 1 \\ 1 & a_1 & 0 & \cdots & 0 \\ 1 & 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & 0 & 0 & \cdots & a_n \end{vmatrix} = \begin{vmatrix} a_0 - \sum_{i=1}^n \frac{1}{a_i} & 1 & 1 & \cdots & 1 \\ 0 & a_1 & 0 & \cdots & 0 \\ 0 & 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & a_n \end{vmatrix}$$

$$= \left(a_0 - \sum_{i=1}^n \frac{1}{a_i} \right) a_1 a_2 \cdots a_n$$

$$\begin{vmatrix}
 x_1 - m & x_2 & x_3 & \cdots & x_n \\
 x_1 & x_2 - m & x_3 & \cdots & x_n \\
 x_1 & x_2 & x_3 - m & \cdots & x_n \\
 \vdots & \vdots & \vdots & & \vdots \\
 x_1 & x_2 & x_3 & \cdots & x_n - m
 \end{vmatrix}
 =
 \begin{vmatrix}
 x_1 - m & x_2 & x_3 & \cdots & x_n \\
 m & -m & 0 & \cdots & 0 \\
 m & 0 & -m & \cdots & 0 \\
 \vdots & \vdots & \vdots & & \vdots \\
 m & 0 & 0 & \cdots & 0 - m
 \end{vmatrix}$$

$$=
 \begin{vmatrix}
 \sum_{i=1}^n x_i - m & x_2 & x_3 & \cdots & x_n \\
 0 & -m & 0 & \cdots & 0 \\
 0 & 0 & -m & \cdots & 0 \\
 \vdots & \vdots & \vdots & & \vdots \\
 0 & 0 & 0 & \cdots & 0 - m
 \end{vmatrix}
 = \left(\sum_{i=1}^n x_i - m \right) (-m)^{n-1}$$

$$\begin{vmatrix}
 x_1 - m & x_2 & x_3 & \cdots & x_n \\
 x_1 & x_2 - m & x_3 & \cdots & x_n \\
 x_1 & x_2 & x_3 - m & \cdots & x_n \\
 \vdots & \vdots & \vdots & & \vdots \\
 x_1 & x_2 & x_3 & \cdots & x_n - m
 \end{vmatrix}
 =
 \begin{vmatrix}
 x_1 - m & x_2 & x_3 & \cdots & x_n \\
 m & -m & 0 & \cdots & 0 \\
 m & 0 & -m & \cdots & 0 \\
 \vdots & \vdots & \vdots & & \vdots \\
 m & 0 & 0 & \cdots & 0 - m
 \end{vmatrix}$$

$$=
 \begin{vmatrix}
 \sum_{i=1}^n x_i - m & x_2 & x_3 & \cdots & x_n \\
 0 & -m & 0 & \cdots & 0 \\
 0 & 0 & -m & \cdots & 0 \\
 \vdots & \vdots & \vdots & & \vdots \\
 0 & 0 & 0 & \cdots & 0 - m
 \end{vmatrix}
 = \left(\sum_{i=1}^n x_i - m \right) (-m)^{n-1}$$

设 $a_1, \dots, a_n \neq 0$.

$$\begin{vmatrix} 1 & 1 & \cdots & 1 & a_0 \\ 0 & 0 & \cdots & a_1 & 1 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & a_{n-1} & \cdots & 0 & 1 \\ a_n & 0 & \cdots & 0 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 1 & \cdots & 1 & a_0 - \sum_{i=1}^n \frac{1}{a_i} \\ 0 & 0 & \cdots & a_1 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & a_{n-1} & \cdots & 0 & 0 \\ a_n & 0 & \cdots & 0 & 0 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 & \cdots & 1 & a_0 - \sum_{i=1}^n \frac{1}{a_i} \\ 0 & 0 & \cdots & a_1 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & a_{n-1} & \cdots & 0 & 0 \\ a_n & 0 & \cdots & 0 & 0 \end{vmatrix} = (-1)^{\frac{n(n+1)}{2}} a_1 a_2 \cdots a_n \left(a_0 - \sum_{i=1}^n \frac{1}{a_i} \right)$$

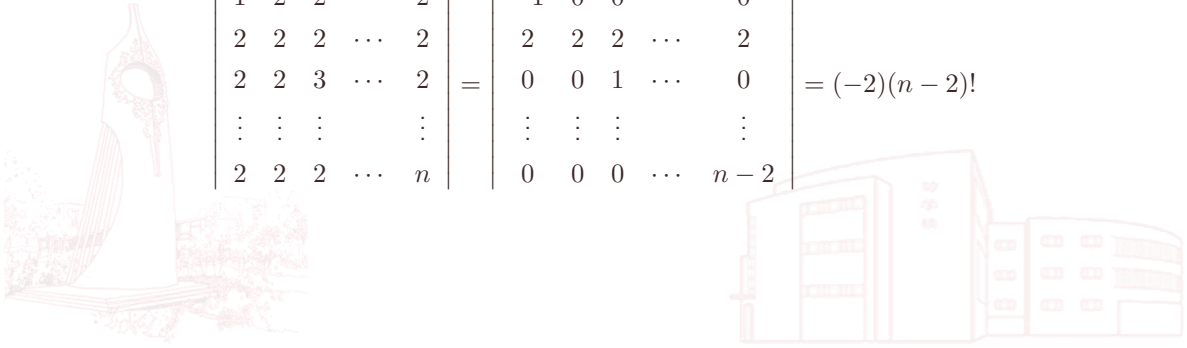
$$\begin{vmatrix} b & \cdots & b & b & a \\ b & \cdots & b & a & b \\ b & \cdots & a & b & b \\ \vdots & & \vdots & \vdots & \vdots \\ a & \cdots & b & b & b \end{vmatrix} = \begin{vmatrix} b & b & \cdots & b & a \\ 0 & 0 & \cdots & a-b & b-a \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & a-b & \cdots & 0 & b-a \\ a-b & 0 & \cdots & 0 & b-a \end{vmatrix}$$

$$= \begin{vmatrix} b & b & \cdots & b & a + (n-1)b \\ 0 & 0 & \cdots & a-b & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & a-b & \cdots & 0 & 0 \\ a-b & 0 & \cdots & 0 & 0 \end{vmatrix}$$

$$= (-1)^{\frac{n(n-1)}{2}} (a + (n-1)b)(a-b)^{n-1}$$

5. 逐行逐列相加减

行列式特点是每相邻两行(列)之间有许多元素相同.用逐行(列)相减可以化出零.


$$\begin{vmatrix} 1 & 2 & 2 & \cdots & 2 \\ 2 & 2 & 2 & \cdots & 2 \\ 2 & 2 & 3 & \cdots & 2 \\ \vdots & \vdots & \vdots & & \vdots \\ 2 & 2 & 2 & \cdots & n \end{vmatrix} = \begin{vmatrix} -1 & 0 & 0 & \cdots & 0 \\ 2 & 2 & 2 & \cdots & 2 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & n-2 \end{vmatrix} = (-2)(n-2)!$$



$$\begin{vmatrix} 1 & 2 & 3 & \cdots & n-1 & n \\ 2 & 2 & 3 & \cdots & n-1 & n \\ 3 & 3 & 3 & \cdots & n-1 & n \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ n-1 & n-1 & n-1 & \cdots & n-1 & n \\ n & n & n & \cdots & n & n \end{vmatrix}$$

$$= \begin{vmatrix} -1 & 0 & 0 & \cdots & 0 & 0 \\ -1 & -1 & 0 & \cdots & 0 & 0 \\ -1 & -1 & -1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ -1 & -1 & -1 & \cdots & -1 & 0 \\ n & n & n & \cdots & n & n \end{vmatrix} = (-1)^{n-1}n.$$


$$\begin{vmatrix}
 1 & 2 & 3 & \cdots & n-1 & n \\
 2 & 3 & 4 & \cdots & n & 1 \\
 3 & 4 & 5 & \cdots & 1 & 2 \\
 \vdots & \vdots & \vdots & & \vdots & \vdots \\
 n-1 & n & 1 & \cdots & n-3 & n-2 \\
 n & 1 & 2 & \cdots & n-2 & n-1
 \end{vmatrix} = \begin{vmatrix}
 1 & 2 & 3 & \cdots & n-1 & n \\
 1 & 1 & 1 & \cdots & 1 & 1-n \\
 1 & 1 & 1 & \cdots & 1-n & 1 \\
 \vdots & \vdots & \vdots & & \vdots & \vdots \\
 1 & 1 & 1-n & \cdots & 1 & 1 \\
 1 & 1-n & 1 & \cdots & 1 & 1
 \end{vmatrix}$$

$$= \begin{vmatrix}
 \frac{n(n+1)}{2} & 2 & 3 & \cdots & n-1 & n \\
 0 & 1 & 1 & \cdots & 1 & 1-n \\
 0 & 1 & 1 & \cdots & 1-n & 1 \\
 \vdots & \vdots & \vdots & & \vdots & \vdots \\
 0 & 1 & 1-n & \cdots & 1 & 1 \\
 0 & 1-n & 1 & \cdots & 1 & 1
 \end{vmatrix}$$

$= \dots\dots$

$$= \frac{n(n+1)}{2} \begin{vmatrix}
 1 & 1 & \cdots & 1 & 1-n \\
 1 & 1 & \cdots & 1-n & 1 \\
 \vdots & \vdots & & \vdots & \vdots \\
 1 & 1-n & \cdots & 1 & 1 \\
 1-n & 1 & \cdots & 1 & 1
 \end{vmatrix}$$

6. 加边法 (添加一行一列, 利于计算, 但同时保持行列式不变)

$$\begin{vmatrix} a+x_1 & a & a & \cdots & a \\ a & a+x_2 & a & \cdots & a \\ a & a & a+x_3 & \cdots & a \\ \vdots & \vdots & \vdots & & \vdots \\ a & a & a & \cdots & a+x_n \end{vmatrix} = \begin{vmatrix} 1 & a & a & a & \cdots & a \\ 0 & a+x_1 & a & a & \cdots & a \\ 0 & a & a+x_2 & a & \cdots & a \\ 0 & a & a & a+x_3 & \cdots & a \\ \vdots & \vdots & \vdots & \vdots & & \vdots \\ 0 & a & a & a & \cdots & a+x_n \end{vmatrix}$$

7.利用数学归纳法证明行列式

证明 (本节设 $\alpha \neq \beta$)

$$\begin{vmatrix} \alpha + \beta & \alpha\beta & 0 & \cdots & 0 & 0 \\ 1 & \alpha + \beta & \alpha\beta & \cdots & 0 & 0 \\ 0 & 1 & \alpha + \beta & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \alpha + \beta & \alpha\beta \\ 0 & 0 & 0 & \cdots & 1 & \alpha + \beta \end{vmatrix} = \frac{\alpha^{n+1} - \beta^{n+1}}{\alpha - \beta}$$

约定零阶行列式 $D_0 = 1$. 当 $n \geq 2$ 时, 将 D_n 按第一行(或第一列)展开得

$$D_n = (\alpha + \beta)D_{n-1} - \alpha\beta D_{n-2},$$

证明

初值验证:

1. 当 $n = 1$ 时, $D_1 = \alpha + \beta$.

根据题设公式, 验证:

$$D_1 = \frac{\alpha^2 - \beta^2}{\alpha - \beta}.$$

利用差平方公式

$\alpha^2 - \beta^2 = (\alpha - \beta)(\alpha + \beta)$, 有:

$$D_1 = \frac{(\alpha - \beta)(\alpha + \beta)}{\alpha - \beta} = \alpha + \beta.$$

所以, $n = 1$ 时结论成立.

2. 当 $n = 2$ 时, $D_2 = (\alpha + \beta)D_1 - \alpha\beta D_0$.

根据题设公式, 验证:

$$D_2 = \frac{\alpha^3 - \beta^3}{\alpha - \beta}.$$

其中, $\alpha^3 - \beta^3 = (\alpha - \beta)(\alpha^2 + \alpha\beta + \beta^2)$,

所以:

$$D_2 = \frac{(\alpha - \beta)(\alpha^2 + \alpha\beta + \beta^2)}{\alpha - \beta} = \alpha^2 + \alpha\beta + \beta^2.$$

因此, $n = 2$ 时结论也成立.

归纳假设： 设 $k \geq 2$ ，假设结论对 $1 \leq n \leq k$ 均成立，特别地：

$$D_k = \frac{\alpha^{k+1} - \beta^{k+1}}{\alpha - \beta}, \quad D_{k-1} = \frac{\alpha^k - \beta^k}{\alpha - \beta}.$$

归纳步骤： 由递推公式及归纳假设，

$$\begin{aligned} D_{k+1} &= (\alpha + \beta)D_k - \alpha\beta D_{k-1} \\ &= \frac{(\alpha + \beta)(\alpha^{k+1} - \beta^{k+1}) - \alpha\beta(\alpha^k - \beta^k)}{\alpha - \beta} \\ &= \frac{\alpha^{k+2} - \beta^{k+2}}{\alpha - \beta}. \end{aligned}$$

因此， $n = k + 1$ 时结论也成立，归纳完成.

8. 利用递推公式

解: 允许 $\alpha = \beta$. 当 $n \geq 2$ 时, 由 $D_n = (\alpha + \beta)D_{n-1} - \alpha\beta D_{n-2}$, 将此式化为:

$$D_n - \alpha D_{n-1} = \beta (D_{n-1} - \alpha D_{n-2}) \quad (1)$$

$$D_n - \beta D_{n-1} = \alpha (D_{n-1} - \beta D_{n-2}) \quad (2)$$

利用公式(1)得:

$$D_n - \alpha D_{n-1} = \beta (D_{n-1} - \alpha D_{n-2}) = \cdots = \beta^{n-2} (D_2 - \alpha D_1) = \beta^n,$$

利用公式(2)得:

$$D_n - \beta D_{n-1} = \alpha (D_{n-1} - \beta D_{n-2}) = \cdots = \alpha^{n-2} (D_2 - \beta D_1) = \alpha^n,$$

不等时联立, 相等时递推, 得
$$D_n = \begin{cases} \frac{\alpha^{n+1} - \beta^{n+1}}{\alpha - \beta}, & \alpha \neq \beta \\ (n+1)\alpha^n, & \alpha = \beta \end{cases}$$

$$D_n = \begin{vmatrix} 7 & 2 & & & \\ 5 & 7 & \ddots & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & 7 & 2 \\ & & & 5 & 7 \end{vmatrix} = 7D_{n-1} - 10D_{n-2}$$



9. 拆项法: 将行列式的其中一行(列)拆成两个数的和

要点: 分解成两个容易求的行列式的和.

$$\begin{aligned}
 D_n &= \begin{vmatrix} a & b & b & \cdots & b & b \\ c & a & b & \cdots & b & b \\ c & c & a & \cdots & b & b \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ c & c & c & \cdots & a & b \\ c & c & c & \cdots & c & a \end{vmatrix} = \begin{vmatrix} c+a-c & b & b & \cdots & b & b \\ c & a & b & \cdots & b & b \\ c & c & a & \cdots & b & b \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ c & c & c & \cdots & a & b \\ c & c & c & \cdots & c & a \end{vmatrix} \\
 &= \begin{vmatrix} c & b & b & \cdots & b & b \\ c & a & b & \cdots & b & b \\ c & c & a & \cdots & b & b \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ c & c & c & \cdots & a & b \\ c & c & c & \cdots & c & a \end{vmatrix} + \begin{vmatrix} a-c & b & b & \cdots & b & b \\ 0 & a & b & \cdots & b & b \\ 0 & c & a & \cdots & b & b \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & c & c & \cdots & a & b \\ 0 & c & c & \cdots & c & a \end{vmatrix}
 \end{aligned}$$